

LUNJIE ZHU

second-year Ph.D. student at HKUST

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EDUCATION & AWARDS

Hong Kong University of Science and Technology

2025-2029 (*expected*)

Ph.D. in Electronic and Computer Engineering

Advisor: Jun Zhang, **IEEE Fellow, Clarivate Highly Cited Researcher**, HKUST Professor

Harbin Institute of Technology, Shenzhen

2021 - 2025

B.Eng. in Information Science and Technology

GPA: 93.615/100 Rank: 7/231

Received **National Scholarship (Top 1%)**, **Outstanding Graduate**, Outstanding Undergraduate Scholarship

Winner of **8** competitions, notably the Third Prize (**Top 12 in the world**) in ICRA Robomaster Sim2Real Challenge

RESEARCH INTERESTS

My research focuses on building efficient and high-performing multimodal generative models. Currently, I explore few-step distillation, autoregressive distillation, and efficient architecture design for multimodal generative models. My prior work spans long-horizon, real-time streaming joint audio-video generation, video VAE decoder acceleration, post-training quantization for large vision-language models, and real-time human novel-view synthesis.

PUBLICATION & PREPRINT

Flash-VAED: Plug-and-Play VAE Decoders for Efficient Video Generation [\[PDF\]](#) [\[Code\]](#)

Lunjie Zhu, Yushi Huang, Xingtong Ge, Yufei Xue, Zhening Liu, Yumeng Zhang, Zehong Lin, Jun Zhang
ICML 2026.

Propose a universal acceleration framework for video VAE decoders that preserves full alignment with the original latent distribution. Extensive experiments on Wan and LTX-Video VAE decoders demonstrate that our method outperforms baselines in both quality and speed, achieving approximately a **6× speedup** while maintaining the reconstruction performance up to **96.9%**. Notably, Flash-VAED accelerates the end-to-end generation pipeline by up to **36%** with negligible quality drops on VBench-2.0.

Omni-LiveAvatar: Minute-Level Real-Time Streaming Joint Audio-Video Avatar Generation [\[Project Page\]](#) [\[Code\]](#)

Lunjie Zhu, Yushi Huang, Xingtong Ge, Yufei Xue, Zhening Liu, Yumeng Zhang, Zehong Lin, Jun Zhang
Submitted.

Proposed **the first** framework for minute-level, real-time streaming joint audio-video avatar generation. Extensive experiments show that Omni-LiveAvatar generates high-quality, synchronized minute-level avatars in real time. In terms of speed, it achieves a **33×** generation speedup over its teacher, LTX-2, on a single NVIDIA H200 GPU; in terms of generation quality, it outperforms accelerated baselines across visual quality, audio quality, cross-modal synchronization, and human fidelity.

Towards Efficient Post-Training Quantization For Large Vision-Language Models [\[PDF\]](#) Via Token-Wise Redundancy Elimination

Yufei Xue, Yushi Huang, Jiawei Shao, **Lunjie Zhu**, Chi Zhang, Xuelong Li, Jun Zhang
Submitted.

Propose a VLM-tailored post-training quantization framework that selectively prioritizes salient tokens while suppressing redundant ones during quantization. Extensive evaluations on 8 benchmarks across 0.5B~32B VLMs demonstrate the state-of-the-art performance of our method, particularly under low-bit settings. For example, it achieves a substantial **16.45%** improvement on MME-RealWorld under 2-bit quantization.

Real-Time Human Frontal View Synthesis from a Single Image [\[PDF\]](#)

Fangyu Lin, Yingdong Hu, **Lunjie Zhu**, Zhening Liu, Yushi Huang, Zehong Lin, Jun Zhang
Submitted.

Proposed PrismMirror, a geometry-guided coarse-to-fine framework for monocular human frontal view synthesis. It employs

a cascade learning strategy to directly extract coarse geometries (SMPL-X meshes and point clouds) before refining textures via rendering supervision. By distilling the unified framework into a lightweight linear attention model, PrismMirror stands as **the first** monocular approach to achieve real-time inference at **24 FPS**, significantly outperforming prior methods in visual authenticity and structural accuracy.

EXPERIENCE

Vivix AI

Latent Star Research Intern

April 2026 - present

Shenzhen, China

- Supervisor: Yi Zhang
- Founded in 2025 and valued at around \$1.32 billion within its first year, Vivix AI is building real-time interactive multimodal AI for live, personalized content creation.
- Research on efficient generative model, focusing on distillation-based acceleration for real-time/long-horizon joint audio-video generation.